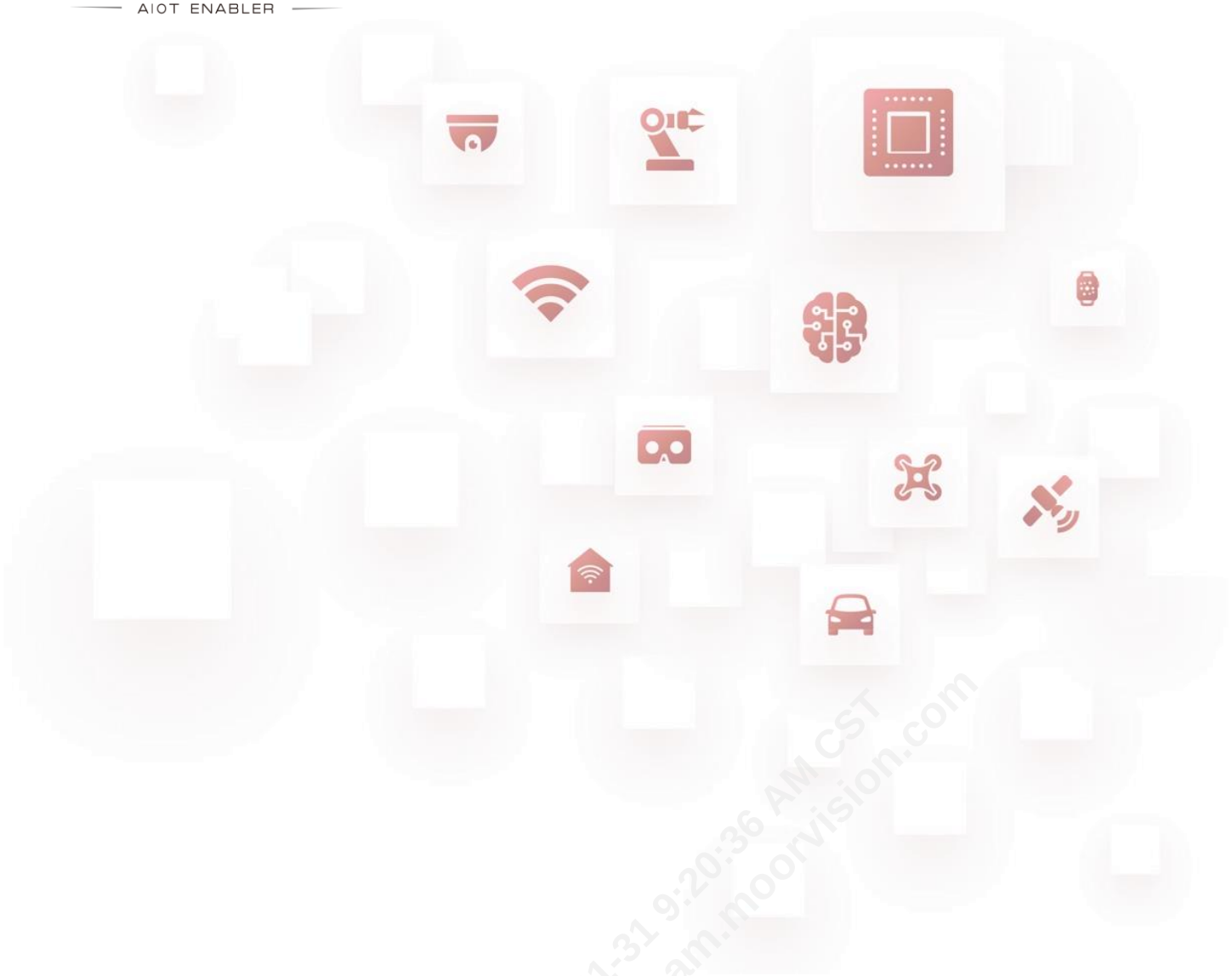
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Thundercomm TurboX Flash for LU

User Guide

Rev. V1.0
Sep 06, 2023

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Revision History

Revision	Date	Author	Description
1.0	Sep 06, 2023	Lining Jiao	Initial release.

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About This Document

- This document describes the usage of Tflash for LU and is applicable to all modules.
- Illustrations in this documentation might look different from your product.
- Depending on the model, some optional accessories, features, and software programs might not be available on your device.
- Depending on the version of operating systems and programs, some user interface instructions might not be applicable to your device.
- Documentation content is subject to change without notice. Thundercomm makes constant improvements on the documentation of the products, including this guidebook.
- Function declarations, function names, type declarations, attributes, and code samples appear in a different format, for example, `cp armcc armcpp`.
- Code variables appear in angle brackets, for example, `<number>`.
- Button, tool, and key names appear in bold font, for example, click **Save** or press **Enter**.
- Commands to be entered appear in a different font; on the host computer use $ as shell prompt, while on the target device use # as shell prompt, for example,

```
$ adb devices
# logcat
```

- Part of the code that does not contain instructions appear in a different format, for example,

```
SUBSYSTEM=="usb", ATTR{idVendor}=="18d1", MODE="0777", GROUP="adm"
```

- Folders, path and files are formatted in italic, for example, *turbox_flash_flat.sh*.

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Chapter 1. Overview

Tflash (TurboX Flash) is an image burning tool developed by Thundercomm. It can be used to flash FlatBuild onto Thundercomm SOM modules.

Tflash includes two versions: Windows graphical interface version and LU command-line version. This document will introduce the usage of the LU command-line version.

Thundercomm releases the LU version of Tflash in the form of deb packages and integrates Tflash into the TurboX SOM SDK Manager (referred to as "sdkmanager") versions 18.04-3.2.0 and 20.04-1.2.0 onwards. Thundercomm has performed comprehensive testing on Tflash within sdkmanager, and thus it is recommended for users to use sdkmanager directly.

FlatBuild is an image format released by Thundercomm, which can be flashed in EDL (Emergency Download) mode. When a device is unable to boot due to software issues, you can repair it by flashing FlatBuild.

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Chapter 2. Install Tflash

2.1. Install sdkmanager (recommended)

Thundercomm has integrated Tflash into sdkmanager and conducted thorough testing. Therefore, it is recommended for users to directly install sdkmanager. For detailed instructions, please refer to: TurboX SOM SDK Manager User Guide.

Users need to install sdkmanager version 18.04-3.2.0, 20.04-1.2.0, or higher. If an older version of sdkmanager is being used, it needs to be upgraded to the new version.

2.2. Install Tflash on PC

Users can also install Tflash on their PC (installed with the LU system) by following the steps below.

2.2.1. Download Tflash installation package

Download address:

https://docs.thundercomm.com/turbox_doc/data/common/tools/thundercomm-tflash-ubuntu.deb

➲ **NOTE:** To download Tflash, you need to log in to https://docs.thundercomm.com. Only Thundercomm staff or users who have purchased Thundercomm products and signed the NDA (Non-Disclosure Agreement) are eligible to access the download.

2.2.2. Install Tflash

Download the Tflash installation package to your local folder and execute the following commands in the terminal of your PC (installed with the LU system):

```
$ sudo dpkg -i Thundercomm-tflash-v1.0.0.deb
```

➲ **NOTE:** "`v1.0.0`" stands for the version number, which will change when a new version is released.

Upon completion of installation, please execute the "`tflash`" command in the terminal. If the following interface appears, it indicates that Tflash has been successfully installed:

```
:~$ tflash

********************************************************************************
Welcome to /usr/bin/tflash version 1.0.0
This script is used to flash FlatBuild and CDT to your device

This script requires root privileges. Please enter the sudo password.

You can obtain the help information via /usr/bin/tflash -h, or get more details form:
https://docs.thundercomm.com/turbox_doc/documents/common/tools/turbox_tflash-user-guide
********************************************************************************


No device found, please connect a device.
```

Figure 2-1. Installation Success

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Chapter 3. Flash with Tflash

3.1. Close serial terminal apps

If the serial terminal app is open, it can lead to failure of flashing. Therefore, it is necessary to close any locally opened serial terminal app before proceeding with the flashing process, such as Minicom on PC.

3.2. Enter 9008 mode

Step 1. Power off the board by disconnecting the power cable and USB cable.

Step 2. Press and hold the **Force_USB_Boot** button (or named "**Force_USB**" on certain development boards).

Step 3. Connect the power cable to the board.

Step 4. Long-press (3 seconds) the **Power** button on the board.

Step 5. Connect the board to PC with a USB Type-C cable (or named "Micro USB cable" on certain development boards).

Step 6. Wait for 3 seconds, and then release the **Force_USB_Boot** button.

Step 7. Execute `lsusb` in the terminal. If the log shows a USB device with the port number **9008**, it indicates a successful connection (see the figure below).

```
:~$ lsusb
Bus 002 Device 003: ID 0bda:9210 Realtek Semiconductor Corp. RTL9210 M.2 NVME Adapter
Bus 002 Device 001: ID 1d6b:0003 Linux Foundation 3.0 root hub
Bus 001 Device 005: ID 04b4:120d Cypress Semiconductor Corp. Convertible 2 TKL
Bus 001 Device 003: ID 046d:c52b Logitech, Inc. Unifying Receiver
Bus 001 Device 120: ID 05c6:9008 Qualcomm, Inc. Gobi Wireless Modem (QDL mode)
Bus 001 Device 014: ID 05e3:0610 Genesys Logic, Inc. Hub
Bus 001 Device 002: ID 05e3:0610 Genesys Logic, Inc. Hub
Bus 001 Device 001: ID 1d6b:0002 Linux Foundation 2.0 root hub
```

Figure 3-1. Execution Result

3.3. Enter FlatBuild directory in the terminal

```
:~/tmp$ cd FlatBuild_Turbox-C610_xx.xx_la2.0.DailyBuild.userdebug.20230829.0147/
:~/tmp/FlatBuild_Turbox-C610_xx.xx_la2.0.DailyBuild.userdebug.20230829.0147$ ls
abl.elf       dspso.bin        logfs_ufs_8mb.bin       prog_firehose_lite.elf   storsec.mbn  Turbox_download_config.xml  userdata_4.img  vbmeta_system.img
aop.mbn       dtbo.img         multi_image.mbn         qdl                      super_1.img  tz.mbn                      userdata_5.img  xbl_config.elf
boot.img      gpt_backup0.bin  NON-HLOS.bin            qupv3fw.elf              super_2.img  uefi_sec.mbn                userdata_6.img  xbl.elf
BTFM.bin      gpt_main0.bin    partition_External.xml  rawprogram0.xml.bak      super_3.img  userdata_10.img             userdata_7.img  xbl_feature_config.elf
cmnlib64.mbn  hyp.mbn          patch0.xml              rawprogram_unsparse0.xml super_4.img  userdata_1.img              userdata_8.img
cmnlib.mbn    imagefv.elf      persist.img             recovery.img             super_5.img  userdata_2.img              userdata_9.img
devcfg.mbn    km4.mbn          prog_firehose_ddr.elf   splash.img               super.img    userdata_3.img              vbmeta.img
```

Figure 3-2. Enter FlatBuild Directory

3.4. Flash with the tflash command

Step 1. Stop ModemManager by executing "`sudo systemctl stop ModemManager`" in the host machine terminal.

Step 2. Execute "`tflash`".

Step 3. When the prompt "`Stop the ModemManager on your host PC and press Enter on your keyboard to continue…`" appears on the screen, press the **Enter** key to proceed with the flashing process.

```
[turbox@sdkmanager-        FlatBuild_Turbox-C610_xx.xx_le2.0.D.userdebug.20230808.1005 ]$ tflash

********************************************************************************
Welcome to /usr/bin/tflash version 1.0.0
This script is used to flash FlatBuild and CDT to your device

This script requires root privileges. Please enter the sudo password.

You can obtain the help information via /usr/bin/tflash -h, or get more details form:
https://docs.thundercomm.com/turbox_doc/documents/common/tools/turbox_tflash-user-guide
********************************************************************************


Image Search Path: /home/turbox/workspace/tmp/FlatBuild_Turbox-C610_xx.xx_le2.0.D.userdebug.20230808.1005
Programmer File: prog_firehose_ddr.elf
Storage Type: eMMC
Image Flash: true
CDT Flash: false
RawProgram Files: rawprogram_unsparse0.xml
Patch Files: patch0.xml
Reset: false
Erase: false
Erase All: false

Stop the ModemManager on your host PC by implementing "sudo systemctl stop ModemManager".
If you require to use the ModemManager, restart it after the flashing process is completed.
The ModemManager can be restarted with the "sudo systemctl start ModemManager" command.
Stop the ModemManager on your host PC and press Enter on your keyboard to continue...
```

Figure 3-3.Run tflash (sdkmanager Screen)

If the following interface is displayed upon completion of tflash operation, it indicates a successful flashing process:

```
16:37:54: INFO:      _             (done)
16:37:54: INFO:     | |
16:37:54: INFO:   __| | ___  _ __   ___
16:37:54: INFO:  / _` |/ _ \| '_ \ / _ \
16:37:54: INFO: | (_| | (_) | | | |  __/
16:37:54: INFO:  \__,_|\___/|_| |_|\___|
16:37:54: INFO: {All Finished Successfully}

16:37:54: INFO: Overall to target 56.045 seconds (42.08 MBps)
16:37:54: INFO: {percent files transferred 100.00%}

Writing log to '/home/jiaoln/tmp/FlatBuild_Turbox-C610_xx.xx_la2.0.DailyBuild.userdebug.20230817.0143/port_trace.txt', might take a minute


Log is '/home/jiaoln/tmp/FlatBuild_Turbox-C610_xx.xx_la2.0.DailyBuild.userdebug.20230817.0143/port_trace.txt'
```

Figure 3-4.Successful Flashing Interface

➲ **NOTE:**

- If you need to restart ModemManager, you can execute "`sudo systemctl start ModemManager`" upon completion of tflash operation.
- When running tflash on PC, you may need to enter your sudo password as prompted. However, when executing tflash within sdkmanager, it is unnecessary to enter the sudo password since it is set to be omitted.

3.5. Restart the board

Step 1. After a successful flashing process, disconnect the USB cable and power cable from the board to power it off.

Step 2. Reconnect the power cable, then press the **Power** button to turn on the board.

➲ **NOTE:** You can confirm if the board has started up correctly by checking the UART log or using adb commands. For specific instructions, please refer to the user manual provided for your specific development board.

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Chapter 4. Additional Usage Methods

The previous chapter introduced the basic usage methods of Tflash. When you execute the tflash command, Tflash will automatically detect the image type and set the relevant parameters.

Alternatively, you can also customize the relevant parameters by command lines when executing the tflash command. To obtain the explanations and usage methods of each parameter, you can execute "`tflash -h`" to access the help documentation.

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